A 28nm 77.2 TFLOPS/W Digital Floating-Point Compute-In-Memory Macro Employing Dynamic Find-Max and Reduced-Cycle Bit-Serial Architecture with Approximation

Jaerok Kim[1], Heeyeon Kim[2] and Yoonmyung Lee[1]

[1] Dept. of Electrical and Computer Engineering, Sungkyunkwan University, Suwon, Korea

[2] Samsung Electronics, Hwaseong, Korea

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Recent advances in Compute-In-Memory (CIM) technology have significantly improved energy efficiency [1-7]. As in Fig. 1, among various data formats, BF16 has been widely adopted due to its exponent-based representation, offering enhanced dynamic range and lower relative error compared to INT's absolute error, thereby reducing accuracy degradation under approximation. However, several critical challenges remain in BF16-CIM macro implementations: (1) synthesis of 8-bit exponent processing circuit demands numerous full adders (FAs) for handling MAC pairs, adversely impacting energy efficiency and throughput [2]; (2) traditional bit-serial operations, although hardware-efficient, typically require multiple cycles (4/8 cycles for 8-bit computations), increasing toggling activity and energy consumption [1,2,6,7]; and (3) existing compact FA structures (e.g., 10T/12T FAs) encounter threshold voltage (V_{TH}) drops, and exact adder trees significantly increase power, latency, and hardware overhead [4,7].

This paper addresses these challenges through three primary innovations: (1) a dynamic find-max (DFMAX) circuit leveraging dynamic logic and masking mechanisms, which reduces area and power consumption by 20.2% and 41.6%, respectively; (2) a novel reduced-cycle bit-serial (RCBS) architecture that efficiently reduces cycle count from 4 to 3 by directly multiplying input sign bits with stored weights and applying LSB approximation. This approach leverages BF16's inherent ability to achieve minimal relative error from LSB approximation, substantially decreasing latency and energy consumption by 55% and 50%, respectively; and (3) alternating 16T/18T FA structures, specifically designed to avoid V_{TH} drop and reduce energy consumption, combined with an approximate adder tree that reduces the required FA count by 20% relative to an exact implementation. Fabricated in a 28nm process, the proposed BF16-CIM macro achieves a peak energy efficiency of 77.16 TFLOPS/W and an area efficiency of 0.324 TFLOPS/mm².

Fig. 2 illustrates the overall architecture and waveform of the proposed CIM macro. The macro consists of a dynamic find-max (DFMAX), a digital shifter & input driver (DSID), a WL driver, a digital shift & adder (DSA), and 8-CIM arrays. Each CIM array includes 4-mantissa weight blocks: the first block stores the sign bit of weights, while the remaining blocks store weights in 2-bit segments. Each weight block consists of 64 sub-arrays and an adder tree, with each sub-array containing four SRAM cells. Through global BL (GBL) and BLB (GBLB), local BL (LBL) and BLB (LBLB) are precharged and then discharged according to the data stored in the SRAM cells, allowing weight values to be read. These values are multiplied with the 2-bit inputs generated by the DSID. The resulting 64 input-weight (IN×W) pairs are summed using an adder tree, and the BF16 MAC results are obtained through the DSA. Because the weights are pre-aligned, the maximum exponent value for shifting is determined using DFMAX from inputs. The inputs are shifted by the DSID according to the difference from the maximum value and then transmitted to the CIM arrays over three cycles, which is the mantissa operation. The exponent and mantissa computations are pipelined, enhancing the throughput of the macro.

Fig. 3 depicts the DFMAX structure and its operational flow. Initially, the system maintains a precharged state. When the start signal becomes high, a delayed enable signal (EN) sequentially activates through delay cells with a delay of t_d. The DFMAX identifies the maximum value bit by bit from MSB to LSB. If at least one input bit among the 64 inputs is high, the MAXB node is discharged, causing the MAX signal to become high. Starting from k=7, when MAX[k]= '1', all EXPs with $EXP_i[k] = 0$ are masked to zero, ensuring these masked bits ($EXP_i[k-1:0]$) do not affect the final outcome. This efficiently determines the maximum among the 64 input pairs. The DFMAX is 20.1% smaller in area and consumes 41.6% less power compared to the synthesis result of digital find-max. The RCBS operation leverages FP properties by incorporating input sign bits directly into SRAM sub-arrays and approximating mantissa LSBs of both inputs and weights. In the proposed architecture, a multiplexer (MUX) is integrated into the SRAM sub-array to select either weight (W) or its complement (WB) based on the input's sign bit, thus efficiently incorporating the sign bit into computations. Leveraging the fact that FP mantissa LSBs have significantly less impact compared to integer LSBs, the proposed method approximates the LSBs of both input and weight. By implementing a 2-bit multiplier within a 3-cycle operation, the toggling of the adder tree is substantially reduced, decreasing latency and energy consumption by 55% and 50%, respectively.

Fig. 4 illustrates the proposed 16T/18T FAs and their schematics and layouts, and structure of the multiplier and adder tree, which significantly impact MAC operations. The 16T occupies a compact area and eliminates V_{TH} drop, ensuring stable operation even at low voltages without optimized transistor sizing. The primary difference between the 16T and 18T FA is the buffering of the summation result. Compared to standard cell FA, 10T and 12T FAs, the proposed 16T achieves lower power consumption; 10T and 12T FAs fail at 0.7V because of the V_{TH} drop. The multiplier implementation varies by column type: the sign column, requiring 2-bit inputs multiplied by a 1-bit weight, is simply realized using only two NOR gates. In contrast, other columns requiring 2-bit by 2-bit multiplication can suffer significant area and power penalties if implemented exactly. To address this, approximation is applied when both input and weight values are '11', resulting in an approximated output of '$1111_{(2)}$'(exact value is '$1001_{(2)}$'). An exact adder tree would require four FAs for 4-bit operations. However, the approximate adder tree employs only three FAs by inserting the OR gate of two inputs' MSBs into the carry-in (C_{IN}), thus minimizing approximation errors and reducing the required number of FAs by 20%. Additionally, C_{IN} signals are buffered, and 16T and 18T FAs alternate at each stage of the adder tree, optimizing overall energy efficiency.

Fig. 5 shows the input-aware toggling rate reduction (IATRR) method and the measurement results of the CIM macro across various supply voltages. The IATRR reduces the toggling rate of FA to hold the input value at cycle2 and cycle3. To minimize error, IATRR is adapted only when the MSBs of cycle2 and cycle3 are the same. A 6.67% energy efficiency increase can be obtained at an input toggling rate of 0.5 This BF16-CIM macro achieves the highest energy efficiency of 77.16 TFLOPS/W at 0.4V (a lower voltage than [4,7]), and the peak area efficiency of 0.324-TFLOPS/mm² at 0.9V. Compared with previous works, this macro achieved the highest energy efficiency. The accuracy drop due to four approximations was evaluated across CIFAR-10 and CIFAR-100 datasets by using AlexNet, ResNet and VGG-Net. The accuracy drop compared to software baseline tested under the conditions of LSB approximation (LSB in Fig.5) and IATRR and adder approximation (LSB & A in Fig.5) are 0.06%, and 0.21%, respectively, using the ResNet-18 model and CIFAR100 dataset. This demonstrates that the hardware gains achieved through approximation significantly outweigh the accuracy losses incurred.

Fig. 6 compares this work with other BF16-CIM papers. Fig. 7 presents the die photo, detailed circuit placement, and chip summary. This BF16-CIM macro, with an area of 0.0576mm², achieved high energy efficiency by using the DFMAX, RCBS, approximate adder tree and IATRR leveraging FP operational characteristics with negligible accuracy loss.

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References:

[1] X. Wang *et al.*, "14.3 A 28nm 17.83-to-62.84TFLOPS/W Broadcast-Alignment Floating-Point CIM Macro with Non-Two's-Complement MAC for CNNs and Transformers," ISSCC, pp. 254-256, 2025.

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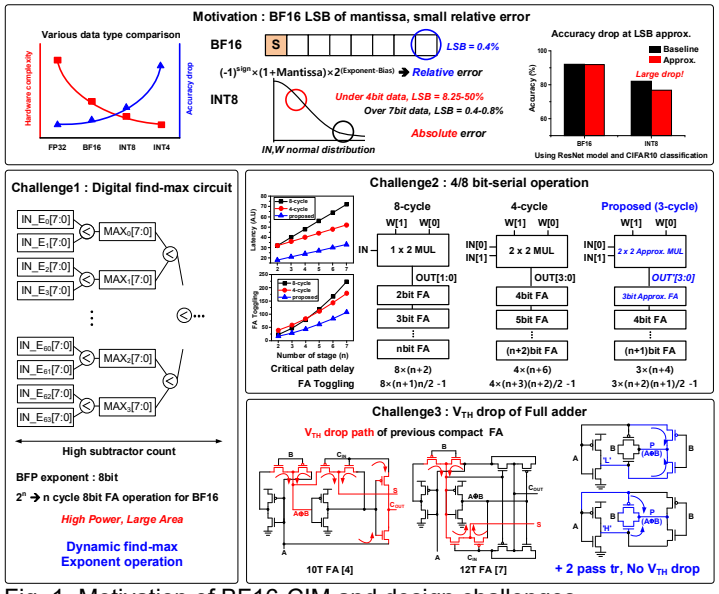


Fig. 1. Motivation of BF16-CIM and design challenges.

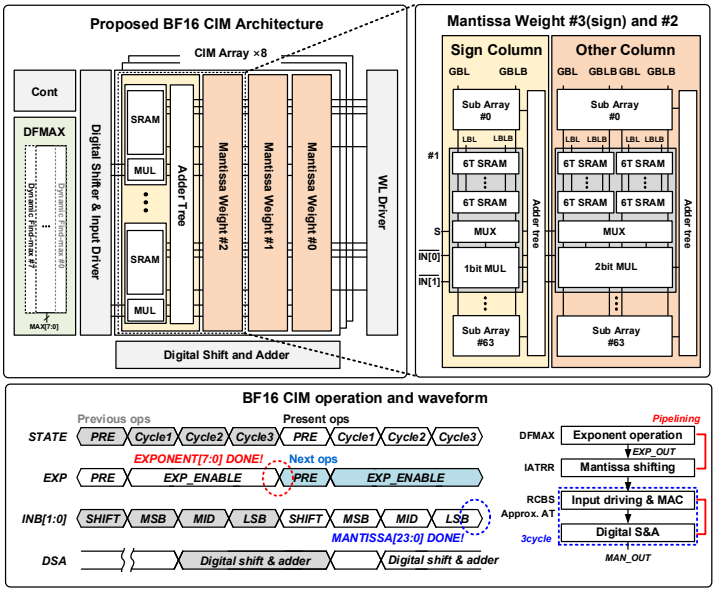


Fig. 2. Overall architecture of CIM macro and waveform.

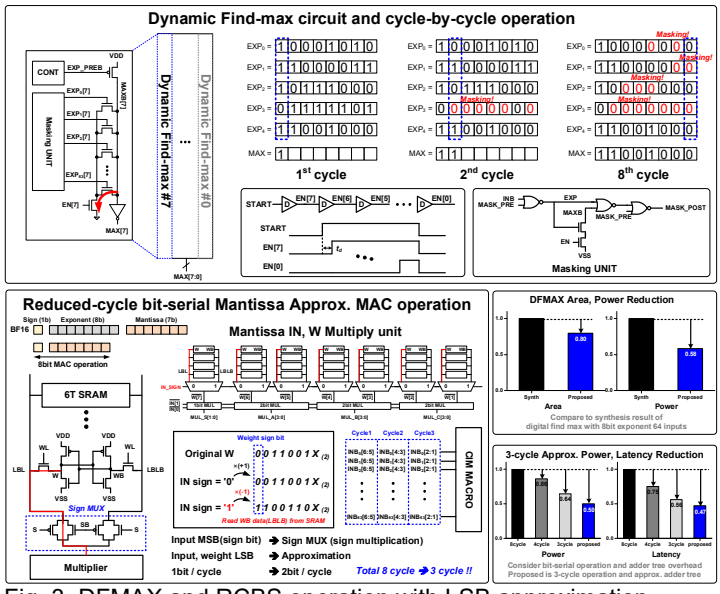


Fig. 3. DFMAX and RCBS operation with LSB approximation.

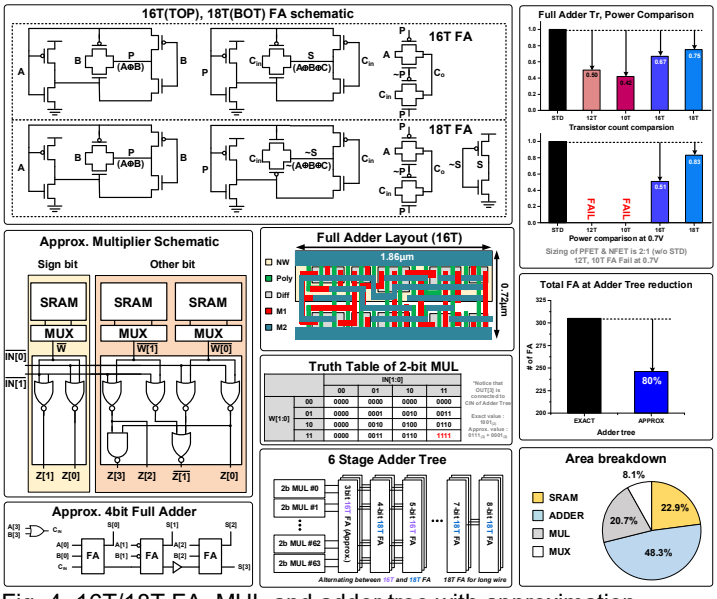
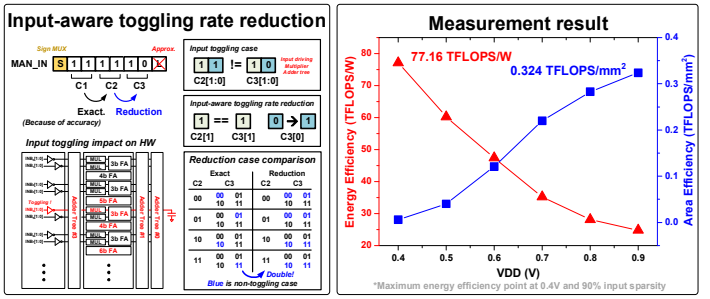
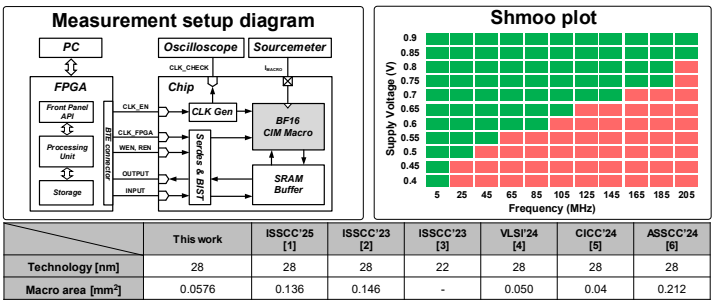


Fig. 4. 16T/18T FA, MUL and adder tree with approximation.



Model		AlexNet	ResNet	VGG16
Dataset		CIFAR10		
Top-1 Error	Baseline	90.55%	96.61%	93.59%
	LSB	90.55%	96.59%	93.55%
	LSB & A	90.40%	96.46%	93.54%
Dataset		CIFAR100		
Top-1 Error	Baseline	65.32%	77.75%	72.71%
	LSB	65.21%	77.69%	72.70%
	LSB & A	65.09%	77.54%	72.64%

Fig. 5. IATRR with approximation, measurement and ML results.

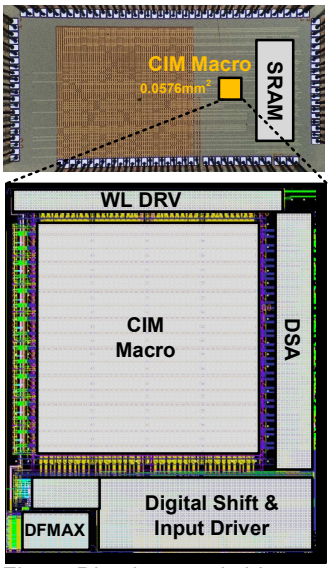


	This work	ISSCC'25 [1]	ISSCC'23 [2]	ISSCC'23 [3]	VLSI'24 [4]	CICC'24 [5]	ASSCC'24 [6]
Technology [nm]	28	28	28	22	28	28	28
Macro area [mm²]	0.0576	0.136	0.146	-	0.050	0.04	0.212
Cell capacity [b]	14k	64k	64k	832k	16k	16k	128k
MAC operation	Digital	Digital	Digital	Hybrid	Digital	Digital	Digital
Approximation	LSB MUL + Adder tree Toggling reduction	-	MUL	-	Adder tree	MUL	MUL Adder tree
Supply voltage [V]	0.4-0.9	0.55-0.9	0.6-0.9	0.6-0.8	0.5-0.9	0.6-1.2	0.6-0.9
Precision	BF16	BF16	BF16	BF16	BF16	BF16	BF16
Area efficiency[1] (TFLOPS/mm²)	0.324	0.697	2.050	-	3.04	1.89	0.929
Energy efficiency[1] (TFLOPS/W)	24.75-77.16	17.83-62.84	14.04-31.6	16.22-17.59	22.47-65.13	12.79-43.06	19.8-45.3
Inference accuracy	77.54 (CIFAR100)	76.13 (ImageNet)	75.94 (CIFAR100)	70.15 (CIFAR100)	91.71 (CIFAR10)	77.36 (CIFAR100)	75.07 (CIFAR100)
Accuracy drop (%)	0.23[2]	0.01	0.17	-	0.46	0.13	1.18

[1]: one operation = multiplication or accumulation and best-number reported
[2]: Maximum accuracy drop at various models and datasets

Fig. 6. Diagram, shmoo plot and comparison table.

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Summary of Chip	
Technology [nm]	28
Macro area [mm²]	0.0576
Cell type	6T SRAM
Cell capacity [b]	14k
MAC type	Digital
Precision	BF16
Supply voltage [V]	0.4-0.9
Approximation	LSB, Adder tree, Toggling
Energy efficiency [TFLOPS/W]	77.16[1]
Throughput [TFLOPS]	0.056[2]
Area efficiency [TFLOPS/mm²]	0.324[2]
Inference accuracy [%, CIFAR10]	96.46[3]
Inference accuracy [%, CIFAR100]	77.54[3]
Maximum accuracy drop [%]	0.23[4]

[1]: input sparsity is 90% and supply voltage is 0.4V
[2]: supply voltage is 0.9V
[3]: Model is ResNet-20
[4]: AlexNet and CIFAR-100 inference case

Fig. 7. Die photo and chip summary.

Additional References:

[2] A. Guo *et al.*, "A 28nm 64-kb 31.6-TFLOPS/W Digital-Domain Floating-Point-Computing-Unit and Double-Bit 6T-SRAM Computing-in-Memory Macro for Floating-Point CNNs," ISSCC, pp. 128-130, 2023.

[3] P. -C. Wu *et al.*, "A 22nm 832Kb Hybrid-Domain Floating-Point SRAM In-Memory-Compute Macro with 16.2-70.2TFLOPS/W for High-Accuracy AI-Edge Devices," ISSCC, pp. 126-128, 2023.

[4] R. Guo *et al.*, "A 28nm 4170-TFLOPS/W/b and 195-TFLOPS/mm2/b Multiply-Free Fully-Digital Floating-Point Compute-In-Memory Macro with Mitchell's Approximation," SOVC, pp. 1-2, 2024.

[5] C. -T. Lin *et al.*, "STAR-SRAM: 43.06-TFLOPS/W, 1.89-TFLOPS/mm2, 400-Kb/mm2 Floating-Point SRAM-Based Digital Computing-in-Memory Macro in 28-nm CMOS," CICC, pp. 1-2, 2024.

[6] A. Guo *et al.*, "A 28nm 128-kb Exponent- and Mantissa-Computation-In-Memory Dual-macro for Floating-point and INT CNNs," A-SSCC, pp 1-3, 2024.

[7] D. Wang *et al.*, "DIMC: 2219TOPS/W 2569F2/b Digital In-Memory Computing Macro in 28nm Based on Approximate Arithmetic Hardware," ISSCC, pp. 266-268, 2022.

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